



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 18 Optimization VI: Convex Optimization

Outline

- Local optimum is also a global optimum in convex optimization
- Examples of convex optimization problems
 - Linear Regression
 - Newsvendor
 - Linear Programming

Local Min $\Rightarrow$ Global Min for a Convex Function

Theorem: Consider the convex optimization problem

$$\min f(x)$$

$$s.t. \ x \in X$$

where $f(x)$ is a convex function and X is a convex set. Then, any **local minimum** is also a **global minimum** 🏆🔥

Minimizing Convex Function vs. Maximizing Concave Function

- If $f(x)$ is *convex* and the feasible set X is *convex*, then any local minimum is a global minimum

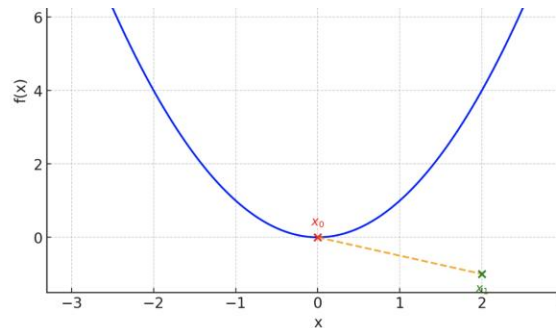
$$\min_{x \in X} f(x) = - \max_{x \in X} (-f(x))$$

- If $f(x)$ is *concave* and the feasible set X is *convex*, then any local maximum is a global maximum

Local Min $\Rightarrow$ Global Min for a Convex Function

Proof (Optional).

- We prove by contradiction. Let $x_0 \in X$ be a local min. Suppose it is not a global min. Then there exists $x_1 \in X$ such that $f(x_1) < f(x_0)$.
- For any $\lambda \in (0,1)$, let $x_\lambda = (1 - \lambda)x_0 + \lambda x_1$. By convexity,
$$f(x_\lambda) \leq (1 - \lambda)f(x_0) + \lambda f(x_1)$$
- Since $f(x_1) < f(x_0)$, the above implies $f(x_\lambda) < f(x_0)$.
- But when $\lambda \rightarrow 0$, $x_\lambda \rightarrow x_0$. This above strict inequality contradicts the local optimality of x_0 !



Examples of Convex Optimization

Example 1: Linear Regression

Recall the Problem Setup

- We are given N samples

$$(X_1, Y_1), \dots, (X_N, Y_N)$$

- The linear model

$$Y_i = \beta_1 X_i + \beta_0 + \epsilon_i$$

- Estimating β_0, β_1 by minimizing the sum of squared errors (SSE)

$$f(\beta_0, \beta_1) = \sum_i (Y_i - \beta_1 X_i - \beta_0)^2$$

Example 1: Linear Regression

- *Exercise: Prove the convexity of SSE using the second-order condition*

Example 1: Linear Regression

- *Exercise: Prove the convexity of SSE using the second-order condition*

Proof. We compute the second-order directional derivative of $f(\beta_0, \beta_1)$ along any direction of (e_0, e_1) as:

- Define $F(t) = \sum_i ((\beta_1 + te_1)X_i + \beta_0 + te_0 - Y_i)^2$
- We have $F'(t) = \sum_i 2((\beta_1 + te_1)X_i + \beta_0 + te_0 - Y_i)(e_1X_i + e_0)$ and

$$F''(t) = 2 \sum_i (e_1X_i + e_0)^2 \geq 0$$

Thus, $f(\beta_0, \beta_1)$ is convex.

Example 2: Newsvendor Problem

- A newsvendor must decide how many newspapers to purchase each morning to maximize expected profit.
 - Demand D is random, with probability density function $f(\cdot)$ and cumulative distribution function $F(\cdot)$.
 - Costs & Revenues
 - Purchase cost per newspaper: c
 - Selling price per newspaper: p
 - Salvage value: 0 (unsold newspapers have no residual value)
 - Decision variable: Order quantity q



Example 2: Newsvendor Problem

- Objective

- Sold quantities: $\min(q, D)$
- Expected revenue: $p \cdot E_D[\min(q, D)]$
- Ordering cost: cq
- Expected profit

$$\pi(q) = E_D[p \min(q, D)] - cq$$

- *Exercise: Show π is concave, or equivalently, $-\pi$ is convex.*
 - Hint: using operations that preserve convexity



Example 2: Newsvendor Problem

- Objective

- Sales: $\min(q, D)$
- Expected revenue: $E_D[p \cdot \min(q, D)]$
- Ordering cost: cq
- Expected profit

$$\pi(q) = E_D[p \min(q, D)] - cq$$

- *Exercise: Show π is concave, or equivalently, $-\pi$ is convex.*
 - Hint: using operations that preserve convexity



Example 2: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
2. The function $h(g(q)) = pg(q)$ is convex in q
3. The expectation $-E_D[p \min(q, D)]$ is convex in q
4. The objective $-\pi(q)$ is convex

Example 2: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
 - Because $g(q) = -\min(q, D) = \max(-q, -D)$ is the maximum of linear functions of q
2. The function $h(g(q)) = pg(q)$ is convex in q
 - Because scaling by a positive constant $p > 0$ preserves convexity
3. The expectation $E_D[p \min(q, D)]$ is convex in q
 - Because weighted sum of convex functions is still convex
4. The objective $-\pi(q)$ is convex
 - Because cq is linear (thus convex), the sum preserves convexity

Example 2: Newsvendor Problem: Finding the Optimal Order Quantity

- Express the objective as an integral

$$\pi(q) = p \int_0^q x f(x) dx + pq(1 - F(q)) - cq.$$

- Setting $\frac{\partial}{\partial q} \pi(q) = 0$:

$$F(q^*) = \frac{p - c}{p}.$$

- Solving for q^* :

$$q^* = F^{-1} \left(\frac{p - c}{p} \right)$$

✅ The optimal order quantity is directly linked to demand uncertainty and price-cost ratio.

🔥 This reinforces why convex optimization is a powerful tool in real-world decision-making! 🚀

Example 3: Linear Programming



Flair Furniture manufactures *tables* and *chairs*

- Objective: Maximize profit while respecting *labor* and *inventory* constraints
- Production requires carpentry and painting labor hours

Item	Carpentry Hours	Painting Hours	Profit per Unit (\$)
Table (T)	3 hours	2 hours	7
Chair (C)	4 hours	1 hour	5

- Resource Constraints
 - Carpentry time available: 2,400 hours
 - Painting time available: 1,000 hours
 - Maximum chairs: 450 (due to inventory limit)
 - Minimum tables: 100 (due to demand requirement)

Example 3: Linear Programming

Exercise: Express the problem as a linear program.



Decision Variables

- T = Number of *tables* produced
- C = Number of *chairs* produced

Example 3: Linear Programming

Exercise: Express the problem as a linear program.



Decision Variables

- T = Number of *tables* produced
- C = Number of *chairs* produced

$$\begin{aligned} &\text{Max } 7T + 5C \\ &s. t. \quad 3T + 4C \leq 2400 \\ &\quad \quad 2T + C \leq 1000 \\ &\quad \quad C \leq 450 \\ &\quad \quad T \geq 100 \\ &\quad \quad T, C \geq 0 \end{aligned}$$

Why is this a Convex Optimization Problem?

Example 3: Linear Programming

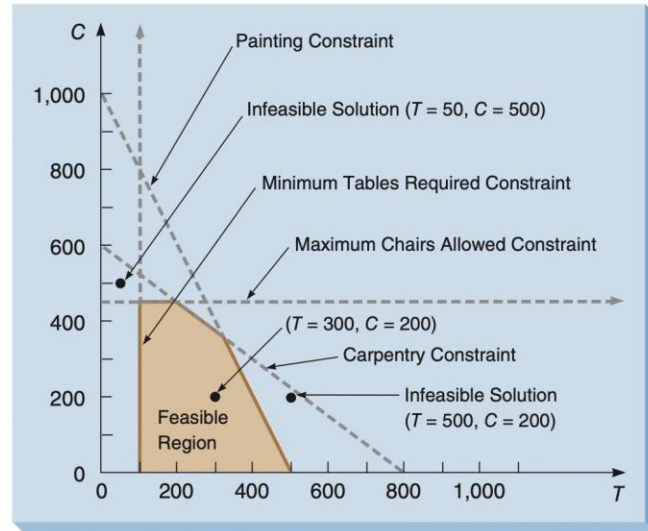
Exercise: Why is this a Convex Optimization Problem?

$$\begin{aligned} & \text{Max } 7T + 5C \\ & s. t. \quad 3T + 4C \leq 2400 \\ & \quad \quad 2T + C \leq 1000 \\ & \quad \quad C \leq 450 \\ & \quad \quad T \geq 100 \\ & \quad \quad T, C \geq 0 \end{aligned}$$

- Objective: linear
- Feasible region: intersection of half spaces

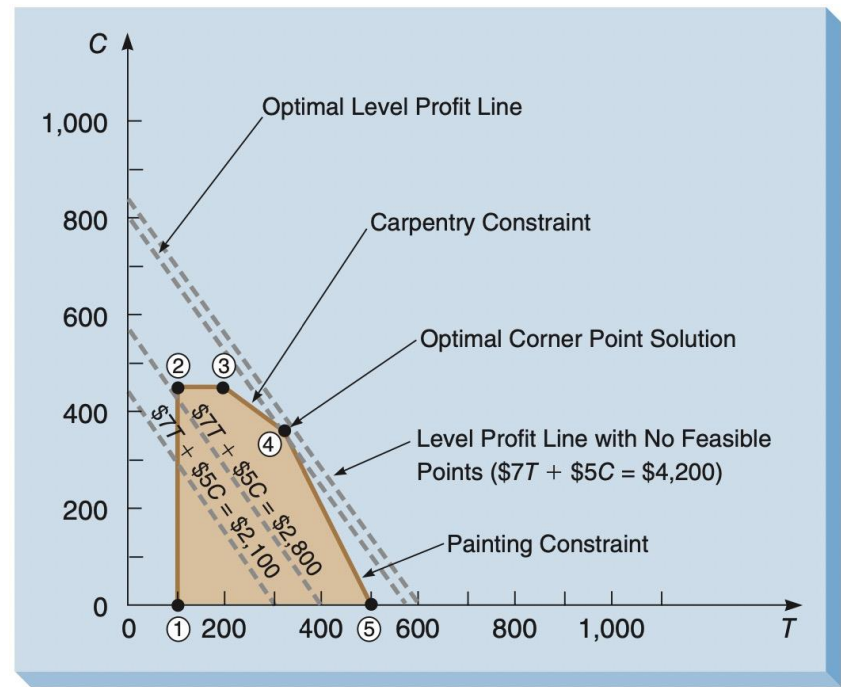
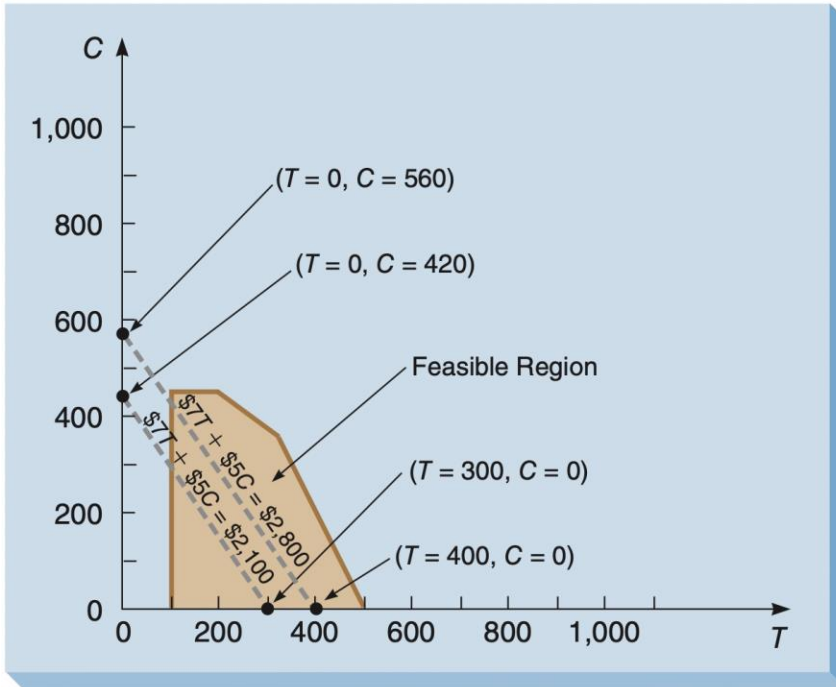
Example 3: Linear Programming

📌 Linear programming problems are convex, which can be efficiently solved using [graphical methods](#), simplex methods, or interior-point methods. 🚀



Example 3: The Linear Programming as a Convex Optimization Problem (4) (Optional)

Graphical methods



Summary

- Proving the convexity
 - Second order condition (e.g. linear regression)
 - Operations that preserve convexity (e.g. newsvendor)
 - Maximum of convex functions
 - Positive scaling
 - Sum
 - Weighted sum (taking expectation)
 - Convexity of feasible region
 - Intersection of half-spaces (e.g. linear programming)
- [Graphical methods](#) for solving linear programming

Midterm Survey:



Your feedback is like the secret ingredient to make this course better! 🍀 Thank you for helping me improve the quality!